WHEN SECTIONS OVER AFFINES ARE COUNTABLE, DERIVED QUASI-COHERENT OBJECTS ARE HOMOTOPY COLIMITS OF PERFECT OBJECTS EXACTLY WHEN THEIR COHOMOLOGY SHEAVES HAVE COUNTABLE SECTION SETS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Homotopy colimits commute with taking cohomology sheaves, and a perfect complex is locally a finite complex of finite free modules. invoked at proof lines 8571-8576. Gives direction (1) implies (2).
- The dga equivalence of the derived category of quasi-coherent modules with the derived category of the dg endomorphism algebra of a perfect K-injective generator, with both functors commuting with homotopy colimits. invoked at proof lines 8583-8601. Transports K to a dg module M over (A, d) and transfers the hocolim statement.
- The countable dga lemma: a dg module whose cohomology groups are countable over a dga with countable cohomology is a homotopy colimit of compact objects. invoked at proof lines 8602-8603. Reduces the converse to countability of the Ext groups.
- The countability of Ext^i(E, E) and Ext^i(E, K) from the section-countability of the cohomology sheaves of K and of O_X. invoked at proof lines 8604-8605. Supplies the countability input for the dga lemma.

Source reference. Original lemma 0CRK: sources/stacks-chow/tags.txt, source lines 16000–16000. Cited at proof lines 8572-8573 for direction (1) implies (2); the countability conclusion is inside the counted span.

Source reference. Original theorem 09IS: sources/stacks-perfect/pinned-original-source.tex, source lines 4680–4680. Cited at proof lines 8583-8585 to transport K to the dg module M; the hocolim transfer is inside the counted span.

Source reference. Original lemma 0CRM: sources/stacks-chow/tags.txt, source lines 16002–16002. Cited at proof lines 8602-8603 to reduce to countability of the Ext groups; the countability inputs are inside the counted span.

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Source reference. Original lemma 0CRP: sources/stacks-perfect/pinned-original-source.tex, source lines 8507–8507. Cited at proof lines 8604-8605 for the countability of Ext^i(E, E) and Ext^i(E, K); the final conclusion of the lemma is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme such that the sets of sections of $\mathcal{O}_X$ over affine opens are countable. Let K be an object of $D_{QCoh}(\mathcal{O}_X)$. The following are equivalent*

1. *$K = hocolim E_n$ with E_n a perfect object of $D(\mathcal{O}_X)$, and*
2. *the cohomology sheaves $H^i(K)$ have countable sets of sections over affine opens.*

Proof. If (1) is true, then (2) is true because homotopy colimits commutes with taking cohomology sheaves (by Derived Categories, Lemma 0CRK) and because a perfect complex is locally isomorphic to a finite complex of finite free $\mathcal{O}_X$-modules and therefore satisfies (2) by assumption on X.

Assume (2). Choose a K-injective complex $\mathcal{K}^\bullet$ representing K. Choose a perfect generator E of $D_{QCoh}(\mathcal{O}_X)$ and represent it by a K-injective complex $\mathcal{I}^\bullet$. According to Theorem 09IS and its proof there is an equivalence of triangulated categories $F : D_{QCoh}(\mathcal{O}_X) \to D(A, \mathrm{d})$ where (A, d) is the differential graded algebra

$$(A, \mathrm{d}) = \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{O}_X)}(\mathcal{I}^\bullet, \mathcal{I}^\bullet)$$

which maps K to the differential graded module

$$M = \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{O}_X)}(\mathcal{I}^\bullet, \mathcal{K}^\bullet)$$

Note that $H^i(A) = \mathrm{Ext}^i(E, E)$ and $H^i(M) = \mathrm{Ext}^i(E, K)$. Moreover, since F is an equivalence it and its quasi-inverse commute with homotopy colimits. Therefore, it suffices to write M as a homotopy colimit of compact objects of $D(A, \mathrm{d})$. By Differential Graded Algebra, Lemma 0CRM it suffices show that $\mathrm{Ext}^i(E, E)$ and $\mathrm{Ext}^i(E, K)$ are countable for each i. This follows from Lemma 0CRP. □